AI-Based Differentiation of

ARDS vs Pulmonary Cardiogenic Edema

Using Lung Ultrasound (Sonography)

A Deep Learning Approach for Sonography-Based Differential Diagnosis

Presented by: **Namburi Achyuth Krishna** | **252CS022** | **MTECH CSE**

Under the Guidance of :

Prof. B. Annappa
Department of Computer Science and Engineering
National Institute of Technology Karnataka

Presentation Agenda

Project Overview — Literature Review Phase

01 **Problem Statement**

Why ARDS vs CPE differentiation is a clinical emergency

02 **The 5 Key LUS Discriminating Features**

Clinical basis for this AI model's learning targets

03 **Literature Review Overview**

14 papers reviewed — structured evidence base

04 **Individual Paper Summaries**

Deep dive into each paper's contribution

05 **Key Insights from Literature**

What we have learned — architecture, features, gaps

06 **Research Gap**

Why this project is novel and needed

07 **Proposed Methodology**

Complete 4-stage project pipeline

08 **Outcomes & Contribution**

Clinical impact and novelty statement

Problem Statement

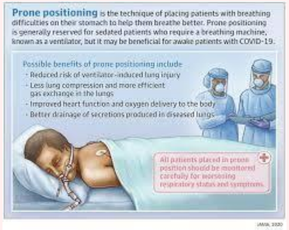
The Clinical Emergency of Misdiagnosis

A patient arrives in the ICU with severe respiratory distress and bilateral B-lines on ultrasound. Is it ARDS or Cardiogenic Pulmonary Edema?

ARDS

Acute Respiratory Distress Syndrome

- ✓ Lung-protective ventilation
- ✓ Treat underlying cause (sepsis/trauma)
- ✓ Prone positioning
- ✓ Steroids / anticoagulation
- ✗ **Diuretics are HARMFUL**



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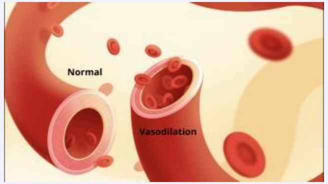
WRONG TREATMENT = DEATH

CPE

Cardiogenic Pulmonary Edema

- ✓ Diuretics (furosemide)
- ✓ Reduce IV fluid
- ✓ Treat heart failure
- ✓ Vasodilators
- ✗ **Lung-protective ventilation unnecessary**

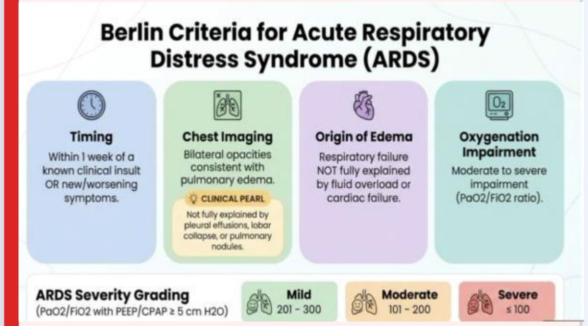


Why Current Methods Fail — Why LUS + AI?

The Diagnostic Gap

Berlin Definition

- Requires CXR / CT scan
- Blood gas analysis needed
- Subjective radiologist read
- Cannot be done bedside fast
- Inter-observer disagreement



Clinical LUS (Human)

- B-lines look IDENTICAL in ARDS & CPE
- Sensitivity only 82–87% (meta-analysis)
- 1 in 7 ARDS cases missed
- Requires trained LUS expert
- Subjective sign interpretation

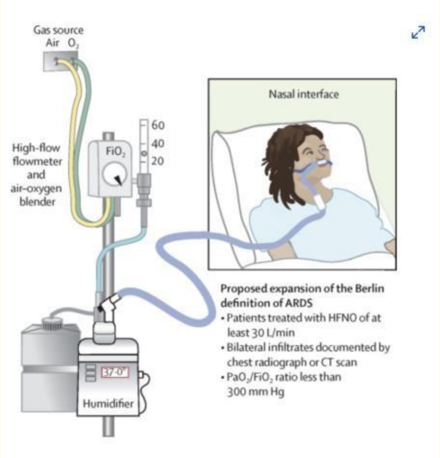
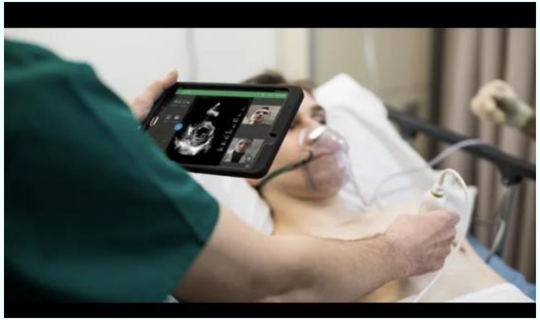
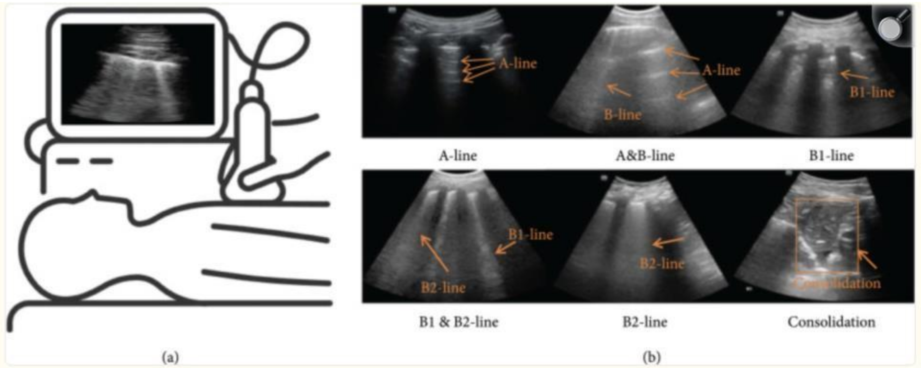


Figure Proposed expansion of the Berlin definition of ARDS

LUS + AI (Current Solution)

- Bedside — real-time, no radiation
- Detects SUBVISUAL pixel patterns
- CNN AUC up to 1.0 on similar tasks
- Objective, reproducible, automated
- Available 24/7 in any setting





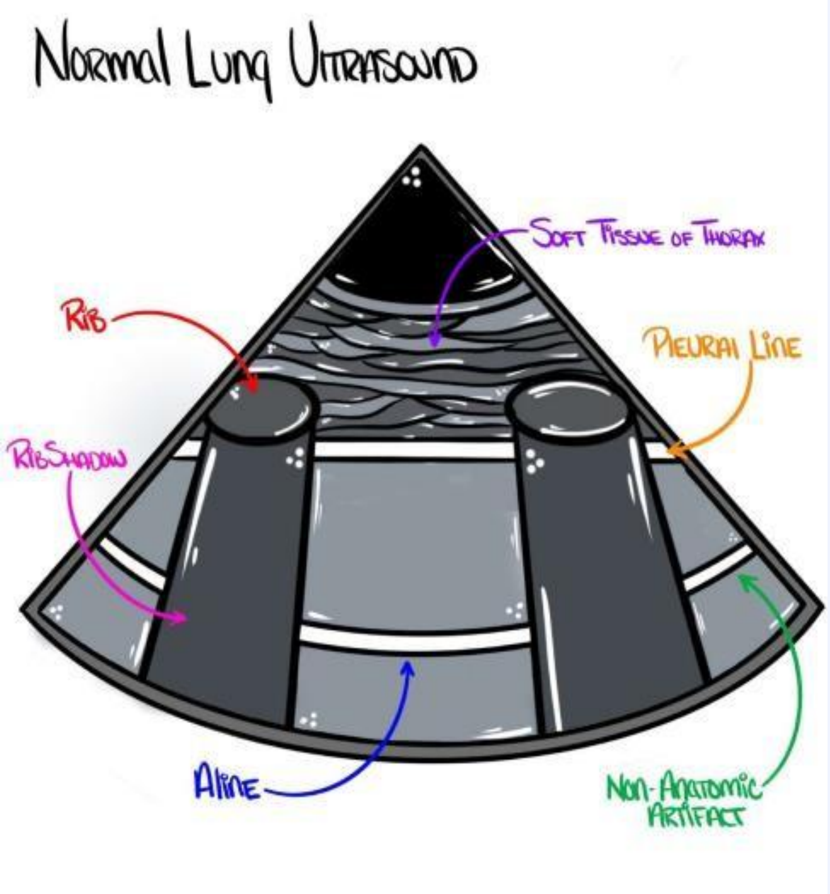
1. **A-lines** (Normal Indicator): Horizontal, echogenic lines parallel to the pleural line.
2. **B-lines** (Pathological Indicator): Vertical, laser-like lines extending from the pleural line to image bottom.
3. **Consolidation Features** (Severe Lung Pathology): Hypoechoic (dark) areas replacing normal air-filled alveoli.

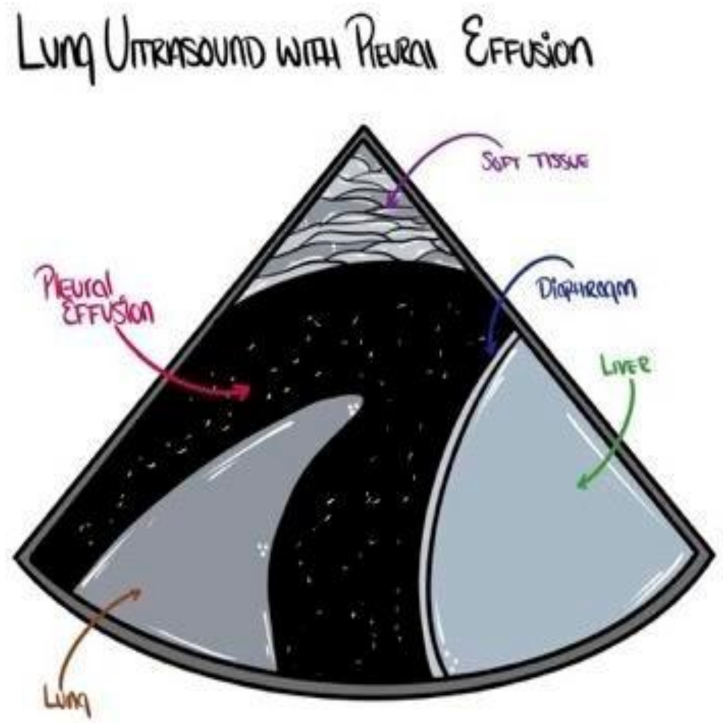
The 5 Key LUS Discriminating Features

What Our AI Model Learns to Detect

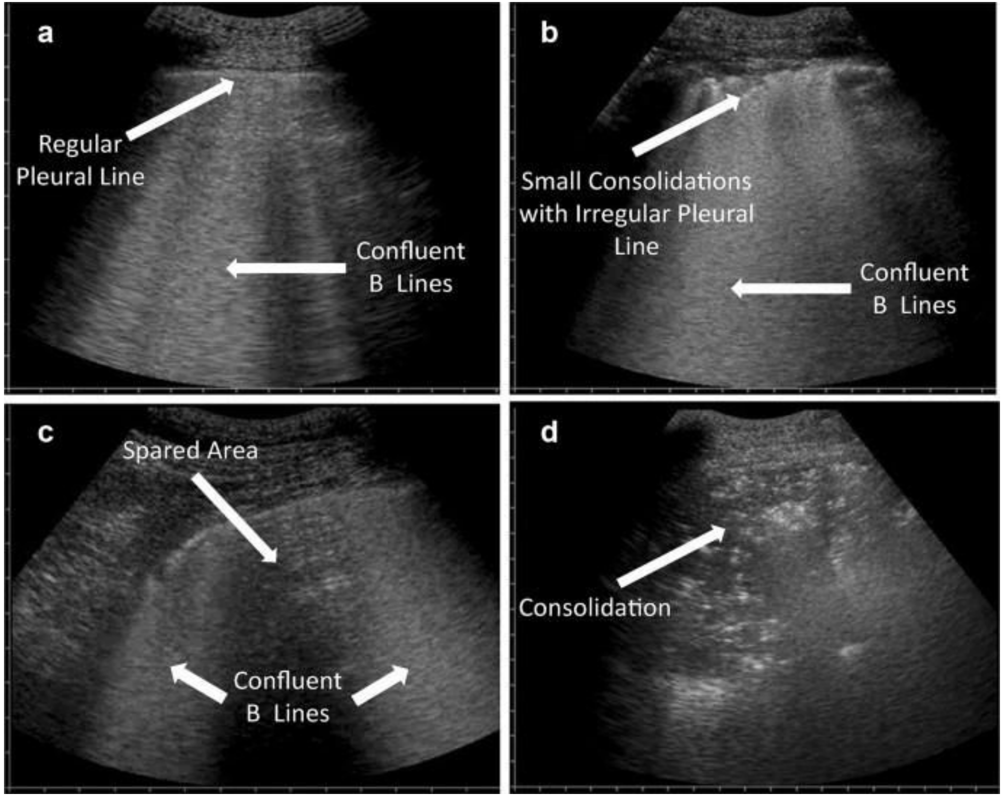
LUS FEATURE	ARDS	Cardiogenic PE
Pleural Line Irregularity	Thick, fragmented, jagged	Thin, smooth, continuous
B-Line Distribution	Heterogeneous, patchy, asymmetric	Homogeneous, bilateral, uniform
Spared Areas	PRESENT — dark gaps between B-lines	ABSENT — uniform B-lines everywhere
Subpleural Consolidations	PRESENT — small wedge-shaped patches	ABSENT — lung structure intact
Lung Sliding	Reduced / Absent in affected zones	Normal / Mildly reduced

Normal Lung vs Lung with B-Lines





Lung Ultrasound with Pleural Effusion
Soft Tissue
Pleural Effusion
Diaphragm
Liver
Lung


a
Regular
Pleural Line
Confluent
B Lines
b
Small Consolidations
with Irregular Pleural
Line
Confluent
B Lines
c
Spared Area
Confluent
B Lines
d
Consolidation

Paper 1 — Arntfield et al. (2021)

CNN to Differentiate Etiology of B-Lines on Lung Ultrasound

PROBLEM STATEMENT

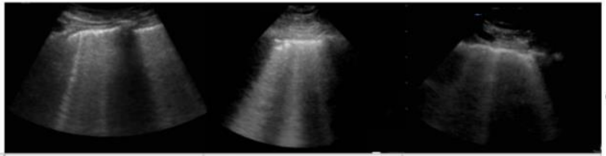
B-lines appear identical in ARDS, COVID ARDS & HPE — physicians cannot distinguish them reliably.

DATASET

243 patients; 612 LUS video clips; 121,381 frames (COVID ARDS, Non-COVID ARDS, HPE)

APPROACH / METHOD

Xception CNN + Transfer Learning (ImageNet). Frame-level classification aggregated to encounter level.



	Hydrostatic Pulmonary Edema	Non-COVID ARDS	COVID-19 ARDS
B Lines	Homogeneous	Patchy	Patchy
Pleural Line	Smooth/Regular	Irregular/Fragmented	Irregular/Fragmented
Sub-pleural consolidation	Absent	Common	Common

EVALUATION METRICS

AUC, Sensitivity, Specificity, F1-score (frame-level & encounter-level)

RESULTS / KEY PERFORMANCE

ENCOUNTER AUC: COVID=1.0 | Non-COVID ARDS=0.934 | HPE=1.0. Outperformed physicians (AUC ~0.789).

CLASSES COMPARED

COVID-19 ARDS vs Non-COVID ARDS vs Hydrostatic Pulmonary Edema

OBSERVATION — HOW THIS GUIDES OUR PROJECT

CORE BASELINE. Proves CNN can differentiate ARDS from pulmonary edema. We use Xception/ResNet + Transfer Learning.

Grad-CAM confirms pleural zone = key ROI.

This project removes COVID confound — pure binary ARDS vs CPE.

LIMITATIONS: Single-centre. Small dataset. COVID may inflate AUC. No external validation.

GAP: No clean binary ARDS vs CPE CNN classifier. This project fills this.

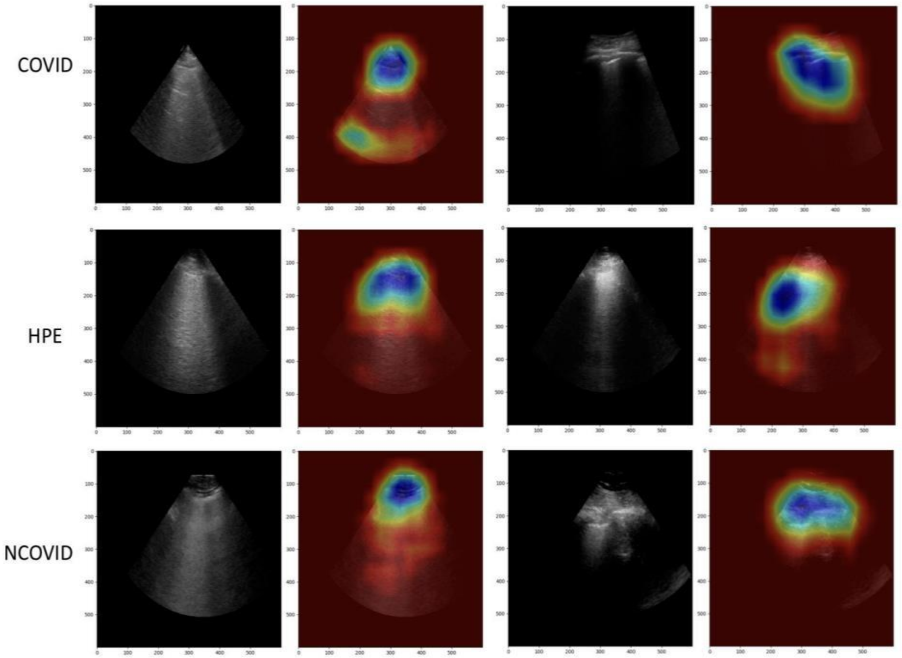


Figure 4 Grad-CAM heatmaps corresponding to a selection of our model's predictions. Blue areas reflect the regions of the image with the highest contribution to the resulting class predicted by the model. In all cases, the immediate area surrounding the pleura appears most activated. COVID, COVID-19 pneumonia; HPE, hydrostatic pulmonary edema; NCOVID, non-COVID-related acute respiratory distress syndrome.

Paper 3 — Chiumello (Campobasso) et al. (2024)

Lung Imaging and Artificial Intelligence in ARDS

PROBLEM STATEMENT

AI applied to ARDS imaging fragmented across labs — no unified comparison of architectures, modalities, or gaps.

APPROACH / METHOD

Review: VGG16, ResNet, InceptionV3, U-Net. Transfer learning. Multi-modality AI comparison.

EVALUATION METRICS

Summary AUC across reviewed models: 0.90–1.00

This review confirms our project fills an identified gap. It guides our architecture selection (ResNet50 or InceptionV3 with transfer learning) and confirms LUS is the optimal modality for our task. The explicit statement that no dedicated ARDS vs CPE binary classifier exists in the literature DIRECTLY VALIDATES OUR PROJECT'S NOVELTY.

RESULTS / KEY PERFORMANCE

ResNet & InceptionV3 + transfer learning: AUC 0.93–1.00. LUS outperforms CXR for bedside AI.

No binary ARDS vs CPE classifier identified.

CLASSES COMPARED

Healthy vs COVID ARDS vs Non-COVID ARDS vs HPE vs Pneumonia (across reviewed studies)

⭐ OBSERVATION — HOW THIS GUIDES TO THIS PROJECT

Directly confirms our project fills an identified gap.

Architecture guide: ResNet50 vs InceptionV3.

Confirms LUS is best modality.

Gap explicitly stated — no binary ARDS vs CPE classifier exists.

LIMITATIONS: Review only. Rapidly evolving field.

GAP: No dedicated binary ARDS vs CPE deep learning classifier. OUR PROJECT.

Paper 6 — Zhao & Bell (2022)

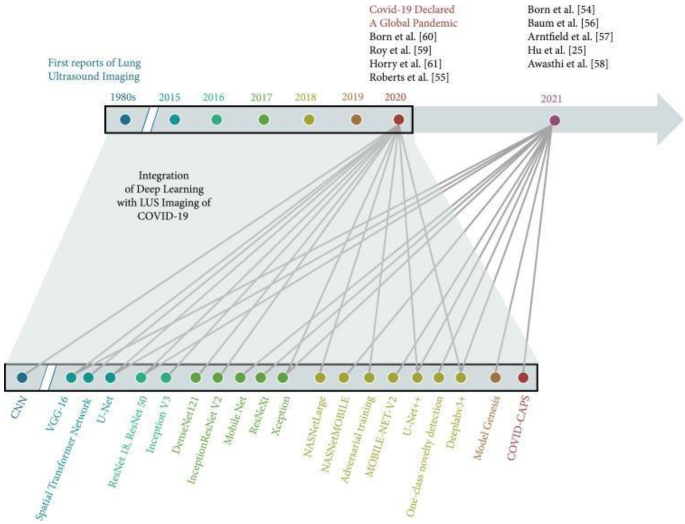
Deep Learning in Lung Ultrasound Imaging of COVID-19 Patients — Review

PROBLEM STATEMENT

20+ DL models built for LUS in isolation — no unified architecture comparison or practical building guide.

APPROACH / METHOD

Review: VGG16, ResNet50, InceptionV3, EfficientNet, CNN+LSTM. Transfer learning & augmentation strategies.



RESULTS / KEY PERFORMANCE

ResNet50 & InceptionV3 best (AUC 0.93–1.00).
CNN+LSTM adds 3–7% AUC for video.
Vertical flip harmful.
Horizontal flip + brightness effective.

CLASSES COMPARED

CPE, ARDS, Pneumonia, COVID-19 vs Normal (across reviewed studies)

OBSERVATION — HOW THIS GUIDES OUR PROJECT

ARCHITECTURE SELECTION GUIDE. We implement **ResNet50 + InceptionV3** with ImageNet transfer learning. CNN+LSTM for lung sliding.
Two-phase fine-tuning protocol. Specific augmentation strategy adopted.

LIMITATIONS: COVID-centric. No unified benchmark. Rapidly evolving.

GAP: No standardised ARDS vs CPE LUS benchmark. This project establishes one.

Key Insights Collected from All 14 Papers

What the Literature Tells Us

Best Architecture

- Xception/ResNet50 + Transfer Learning — core model (Paper 1)
- InceptionV3 — best for multi-scale B-line patterns (Papers 3,6)
- CNN+LSTM — adds 3–7% AUC for video / lung sliding (Papers 6,9)
- Two-phase fine-tuning: freeze → unfreeze last blocks (Paper 9)

Best Features / ROI

- Pleural line zone = primary CNN attention (Grad-CAM, Paper 1)
- Heterogeneous B-lines = strongest single discriminator (Papers 2,13,7)
- Spared areas = highest specificity for ARDS (Papers 5,7,14)
- Anterolateral 12 zones = priority input regions (Paper 13)

Evaluation Design

- Head-to-head: CNN vs LUS-trained physicians (Papers 1,4)
- Report: AUC + sensitivity + specificity (Papers 4,7)
- Baseline to beat: human AUC 0.89–0.93 (Paper 7)
- Focal vs Non-Focal ARDS subgroup analysis (Papers 11,7)

Research Gaps Confirmed

- No binary ARDS vs CPE CNN classifier confirmed (Paper 3)
- No temporal lung sliding model exists (Papers 3,6)
- Human sensitivity only 82–87% — 1 in 7 missed (Paper 7)
- GLCM classical ML as baseline comparison available (Paper 2)

Research Gap — Why Our Project is Needed

What the Literature is Missing

A 2024 comprehensive review (Chiumello et al.) explicitly confirms: No dedicated binary ARDS vs CPE deep learning classifier on LUS exists in the literature.

GAP 1

No Binary ARDS vs CPE CNN

All DL studies bundle COVID, use multi-class, or compare vs Normal — never pure ARDS vs CPE

Papers 1, 3, 4

GAP 2

No Temporal Lung Sliding Model

No study has used CNN+LSTM specifically to capture lung sliding for ARDS vs CPE

Papers 3, 6, 9

GAP 3

Human Sensitivity = Only 82–87%

1 in 7 ARDS cases misdiagnosed even by expert LUS. AI can close this gap.

Paper 7

GAP 4

No Multi-Centre Validated Model

All existing LUS AI models are single-centre. External validation is absent.

Papers 1, 3, 6

GAP 5

Explainability Not Correlated with Signs

Grad-CAM used but never systematically correlated with the 5 clinical LUS signs.

Papers 1, 3

OUR PROJECT

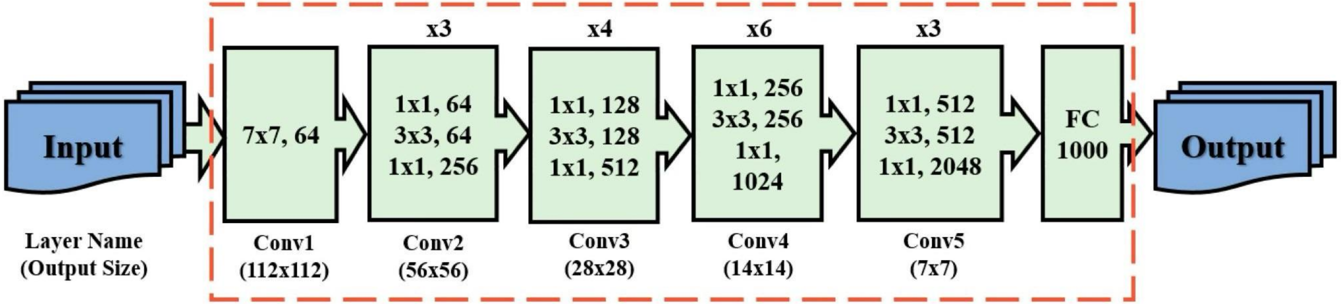
Addresses ALL 5 Gaps

Binary ARDS vs CPE | CNN+LSTM | Improve sensitivity | Multi-centre design | Grad-CAM + sign correlation

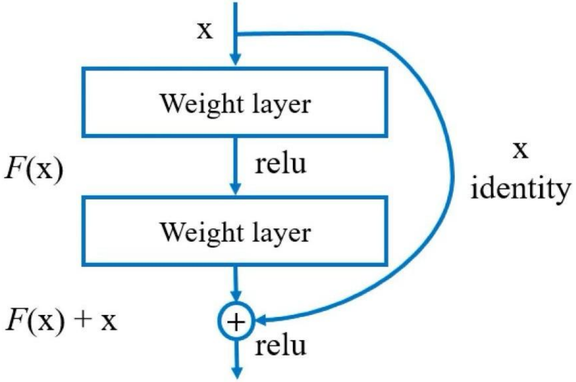
Novel contribution

Model 1

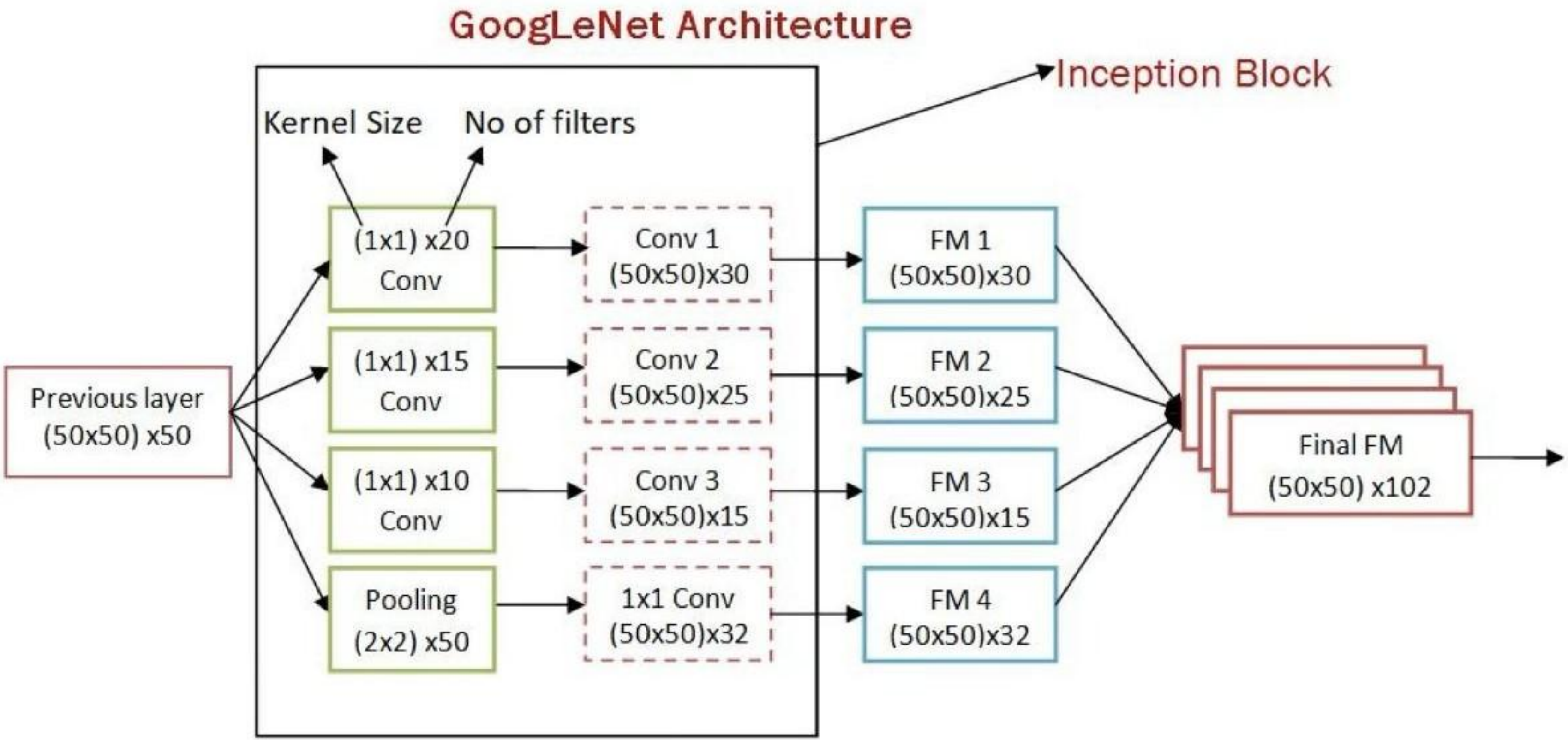
A. ResNet 50 Diagram



B. Residual learning

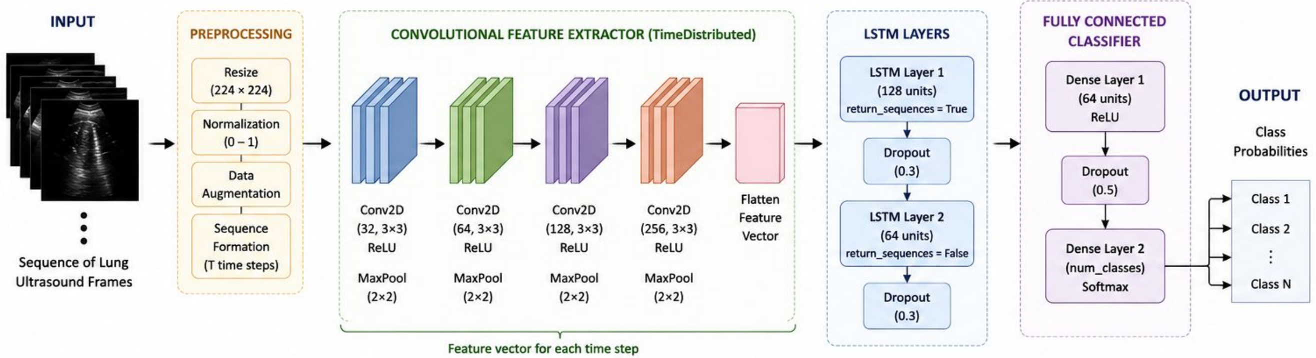


Model 2


GoogLeNet Architecture
Inception Block
Kernel Size
No of filters
Previous layer
(50x50) x50
(1x1) x20
Conv
(1x1) x15
Conv
(1x1) x10
Conv
Pooling
(2x2) x50
Conv 1
(50x50)x30
Conv 2
(50x50)x25
Conv 3
(50x50)x15
1x1 Conv
(50x50)x32
FM 1
(50x50)x30
FM 2
(50x50)x25
FM 3
(50x50)x15
FM 4
(50x50)x32
Final FM
(50x50) x102

Model 3

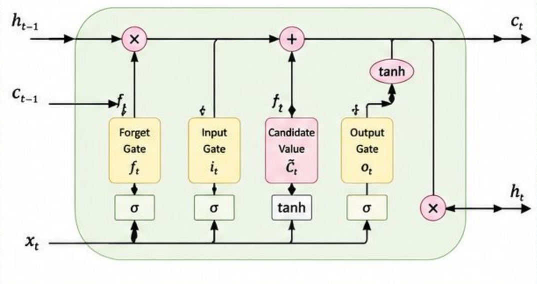
LSTM ARCHITECTURE
FOR PULMONARY ULTRASOUND CLASSIFICATION



ARCHITECTURE SUMMARY

Stage	Configuration
Input	Sequence of T frames, each of size 224 × 224 × 1 (or 3)
Conv Block 1	Conv2D (32, 3×3) + ReLU + MaxPool (2×2)
Conv Block 2	Conv2D (64, 3×3) + ReLU + MaxPool (2×2)
Conv Block 3	Conv2D (128, 3×3) + ReLU + MaxPool (2×2)
Conv Block 4	Conv2D (256, 3×3) + ReLU + MaxPool (2×2)
Flatten (per time step)	Feature vector
LSTM Layer 1	128 units, return_sequences = True
LSTM Layer 2	64 units, return_sequences = False
Dense Layer	64 units, ReLU
Output Layer	num_classes, Softmax

LSTM CELL STRUCTURE



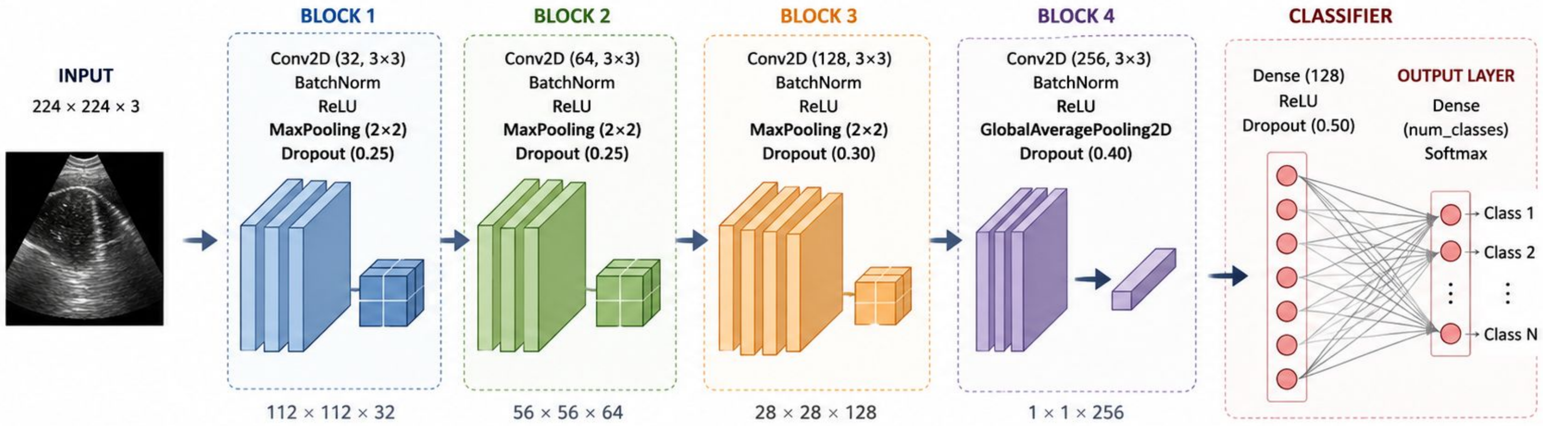
NOTATIONS

- x_t : Input at time step t
- h_t : Hidden state at time step t
- c_t : Cell state at time step t
- σ : Sigmoid activation
- tanh : Hyperbolic tangent
- × : Element-wise multiplication
- + : Element-wise addition

This LSTM architecture captures spatio-temporal information from a sequence of lung ultrasound frames by combining CNN-based feature extraction with LSTM's temporal modeling capability, making it suitable for pulmonary pattern classification.

PROPOSED ARCHITECTURE (NOVELTY)

Proposed Custom CNN Architecture



- Conv2D + BatchNorm + ReLU
- MaxPooling (2×2)
- Global Average Pooling
- Fully Connected Layer
- Output (Softmax)

Block	Conv Filters	Kernel Size	Output Size	Dropout
Block 1	32	3 × 3	112 × 112 × 32	0.25
Block 2	64	3 × 3	56 × 56 × 64	0.25
Block 3	128	3 × 3	28 × 28 × 128	0.30
Block 4	256	3 × 3	1 × 1 × 256 (GAP)	0.40
Dense	128	–	128	0.50
Output Layer	num_classes	–	num_classes	–

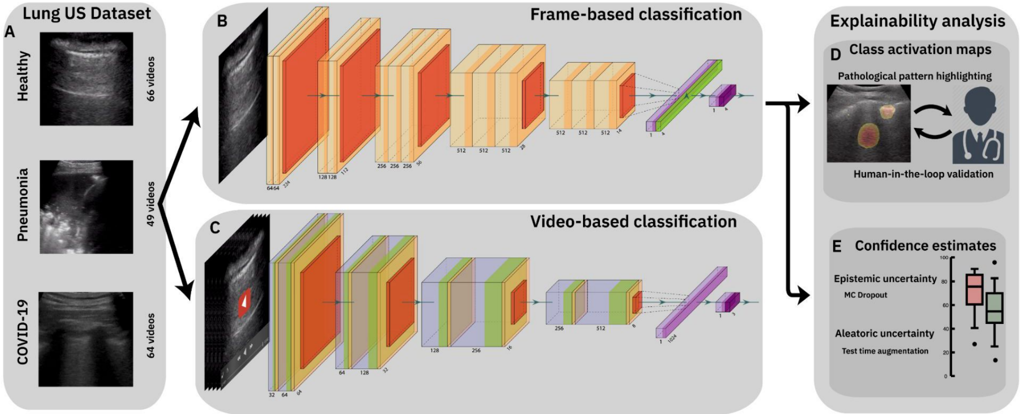
ARCHITECTURE SUMMARY

- Lightweight and optimized for lung ultrasound images
- Batch Normalization for stable training
- Dropout for regularization
- Global Average Pooling to reduce parameters and overfitting
- Softmax output for multi-class classification

Dataset Details

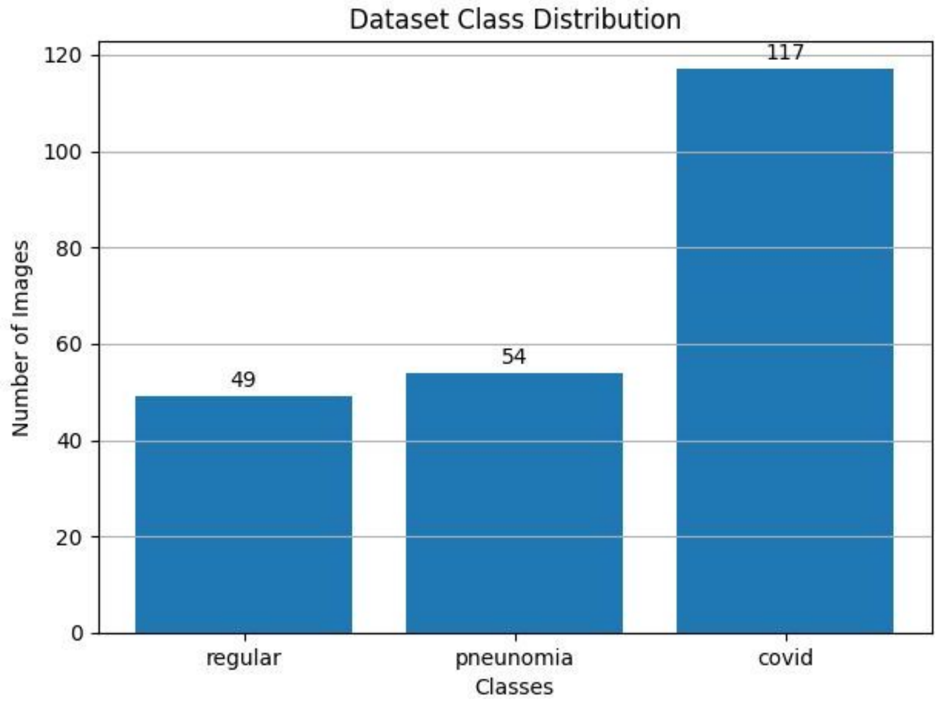
COVID-19 Lung Ultrasound Dataset

- Source: GitHub – COVID-19 Lung Ultrasound Dataset
- Type: Lung Ultrasound (LUS) Images & videos
- Classes available:
  - COVID / Viral Pneumonia
  - Bacterial Pneumonia
  - Healthy
- Probe types:
  - Linear probe
  - Convex probe
- Characteristics:
  - Real clinical ultrasound scans
  - Includes artifacts (B-lines, consolidations)
- Limitation:
  - Not specifically labeled as ARDS vs PCE
  - Used as **proxy dataset for initial model validation**



Overview figure about current efforts. Public dataset consists of >200 LUS videos.

Dataset Details



Sample Images from Dataset

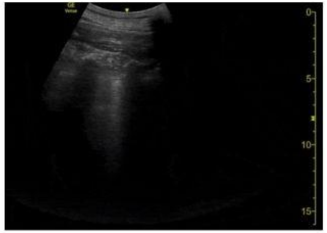
regular

regular

pneunomia

pneunomia

covid



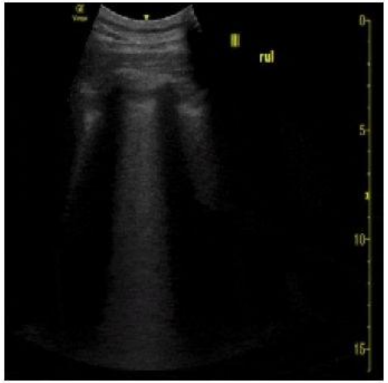
covid

Preprocessing Pipeline–

- Image resizing:
  - ResNet → 224×224
  - Inception → 299×299
- Grayscale → Converted to 3-channel (if needed)
- Normalization:
  - Pixel scaling (0–1)
- Data augmentation:
  - Rotation (±15°)
  - Horizontal flip
  - Zoom
- Noise handling:
  - Ultrasound speckle preserved (important feature)
- Dataset split:
  - Train: 70%
  - Validation: 15%
  - Test: 15%
  - Patient-level split (no leakage)

Model Output

True: covid
Pred: covid



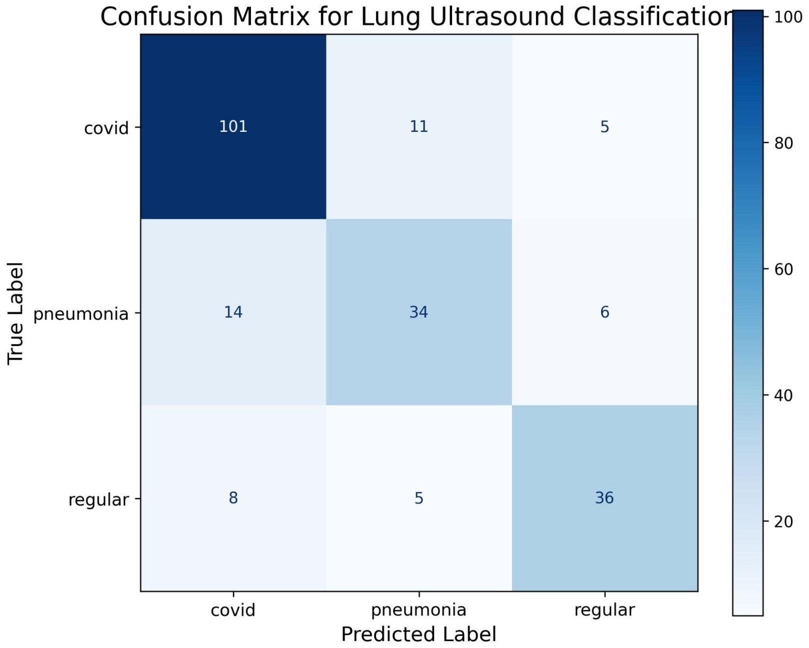
True: covid
Pred: covid

True: covid
Pred: covid

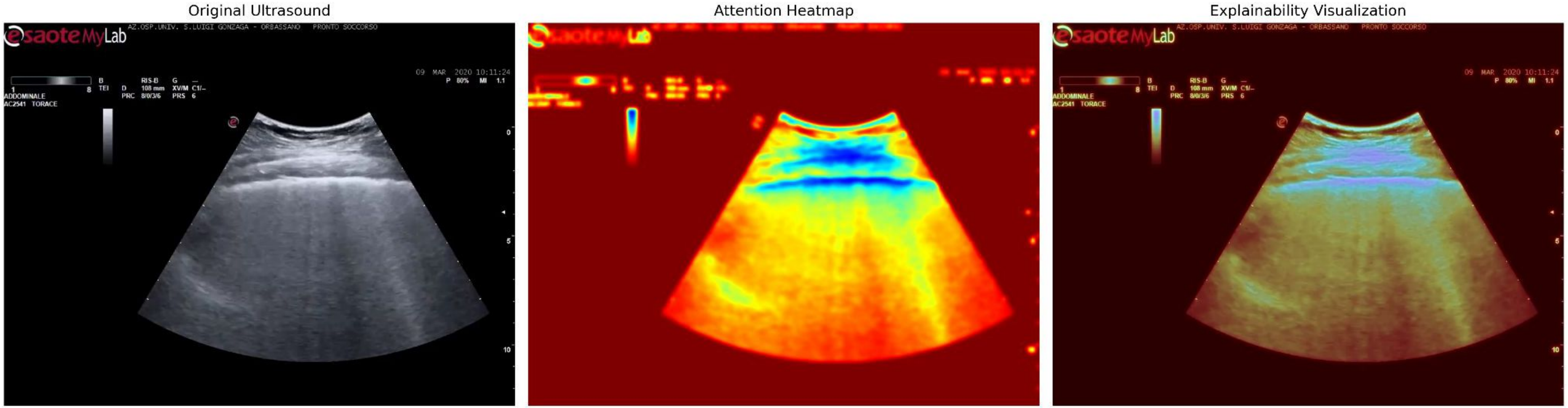
True: covid
Pred: covid

True: regular
Pred: covid

Model Results

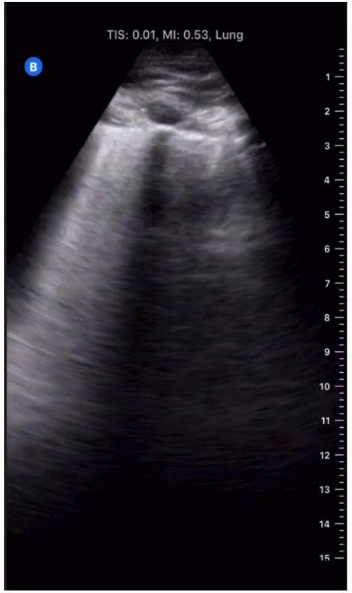


Grad-CAM Explainability (NOVELTY)

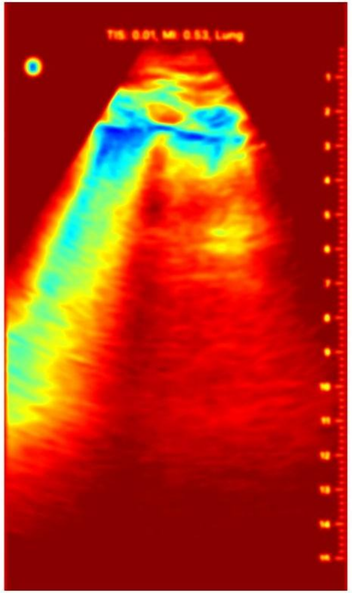


Grad-CAM Explainability (NOVELTY)

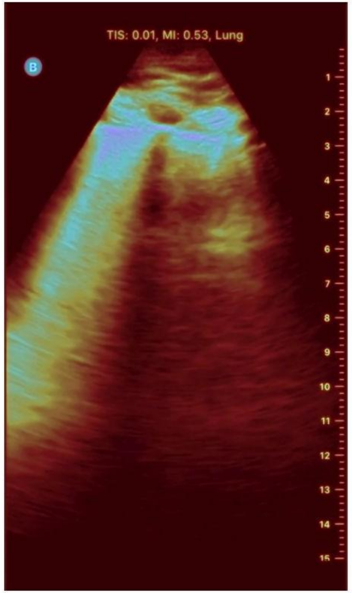
Original Ultrasound



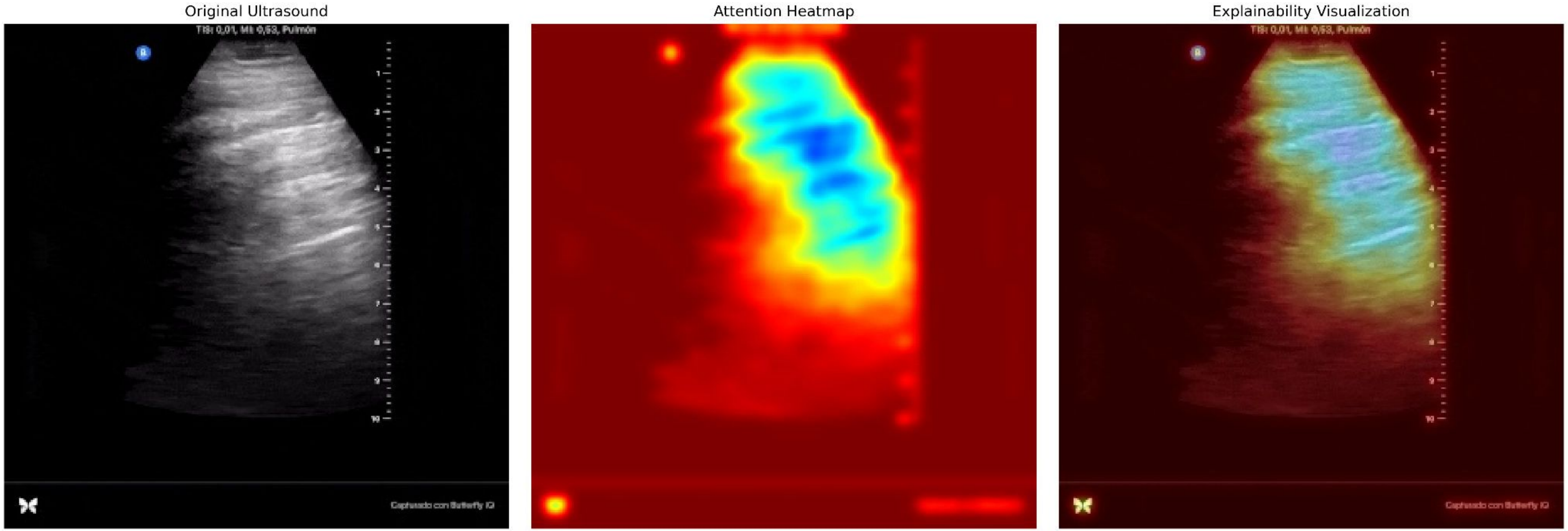
Attention Heatmap



Explainability Visualization



Grad-CAM Explainability (NOVELTY)



Thank You